DermXAI: Hospital Clinical Decision Support System

Official Technical Documentation • B.Tech CSE Final Year Capstone Project

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Project Architectural Overview & Scope

DermXAI is an end-to-end Healthcare & Dermatopathology Decision Support System (DSS). It synchronizes patient ABCDE skin lesion evaluations with active hospital operations, featuring Role-Based Access Control (Admin, Doctor, Pharmacy, Lab, Patient), administrative approval for all registrations, client-side vector PDF generation, and automated multi-department order routing.

Section 2: A to Z Codebase Guide (What Every File & Code Does)

`frontend/app/page.jsx`— Hospital Landing Page & Portal Gateway

Public entry point containing the navigation bar, emergency care cards, hospital room availability, shortcut to the 3D Anatomical Health Radar, and routing to Sign In and Sign Up.

`frontend/app/layout.jsx`— Root Application Shell & Font Provider

Applies global viewport configurations, initializes GeistSans medical font family, and wraps all pages in the root Next.js 16 App Router structure.

`frontend/app/globals.css`— Global Styling & Tailwind CSS Theme

Defines the clean hospital white and slate color system (#306CE9 medical blue, neutral slate borders), scrollbar formatting, and glassmorphism panel styles.

`frontend/app/(auth)/login/page.jsx`— Hospital Portal Authentication Desk

Authenticates all 5 hospital roles. Enforces password checks, verifies Admin approval status on accounts (blocks unapproved staff), and provides quick-fill evaluation chips.

`frontend/app/(auth)/register/page.jsx`— Multi-Role User & Staff Registration

Allows real patients and staff to register with their own name, email, password, and specialty/license. Sets all first-time sign-ups to PENDING for Admin verification.

`frontend/app/(dashboard)/layout.jsx`— Global Dashboard Shell & Protected Route Guard

Guards all /admin, /doctor, /medical, /lab, and /patient routes. Redirects unauthorized users to /login, displays user role identity badges, and manages secure sign-out.

`frontend/app/(dashboard)/(patient)/patient/dashboard/page.jsx` — Patient Diagnostic Portal & AI Scanner

Enables patients to upload skin lesion dermoscopy images, tag anatomical body regions on a 3D model, view clinical ABCDE criteria, and route cases to the doctor.

`frontend/app/(dashboard)/(patient)/patient/records/page.jsx` — Patient Electronic Health Records (EHR)

Comprehensive patient portal with 3 tabs: Physician Consultations (with downloadable Vector PDF report), Pharmacy Prescription fulfillment, and Pathology Lab biopsy findings.

`frontend/app/(dashboard)/(patient)/patient/appointments/page.jsx` — Patient Appointments Schedule

Displays upcoming and historical clinical consultations, assigned specialist doctor names, consultation suite rooms, and appointment time slots.

`frontend/app/(dashboard)/(doctor)/doctor/dashboard/page.jsx` — Attending Physician Clinical Desk & Triage

Displays incoming patient dermoscopy evaluations. Features interactive modals for the doctor to prescribe custom medicines (dispatched to Pharmacy) and order punch biopsies (dispatched to Lab).

`frontend/app/(dashboard)/(doctor)/doctor/patients/page.jsx` — Hospital Patient Registry Directory

Searchable clinical database of all patients undergoing automated AI dermoscopy and active specialist follow-ups.

`frontend/app/(dashboard)/(doctor)/doctor/xai-analysis/page.jsx` — Standalone XAI Lesion Classifier

Physician tool for direct image upload, deep learning inference, Grad-CAM heatmap visualization, and differential pathology research dossier review.

`frontend/app/(dashboard)/(admin)/admin/dashboard/page.jsx` — Hospital Admin Operations Tower

Supervised by waheeddeveloper@gmail.com. Tracks hospital bed capacity, operational KPIs, pending staff approvals count, and patient appointment room dispatch.

`frontend/app/(dashboard)/(admin)/admin/users/page.jsx` — Central Staff Registry & Approval Control Desk

Exclusive Admin console featuring the "Registrations Pending Admin Approval" panel. Admin reviews and approves or rejects newly registered Doctors, Patients, Pharmacists, and Lab Specialists.

`frontend/app/(dashboard)/(medical)/medical/dashboard/page.jsx` — Clinical Pharmacy & Dispensary Desk

Receives live e-prescriptions dispatched by doctors. Allows pharmacists to verify formulations, record safety counseling, dispense medications, and print pharmacy labels.

`frontend/app/(dashboard)/(lab)/lab/dashboard/page.jsx` — Dermatopathology & Tissue Biopsy Lab Desk

Receives punch biopsy orders from doctors. Technicians select tissue stains (Fontana-Masson, H&E), assess microscopic surgical margins, and release finalized pathology certificates.

`frontend/app/test-3d/page.jsx` — 3D Anatomical Lesion Radar

Interactive 3D mesh model allowing patients and physicians to orbit, pan, and mark the spatial anatomical location of cutaneous lesions.

`frontend/lib/hospitalStore.js` — Hospital Central State & Collaboration Engine

The central backbone synchronizing live queues between Patients, Doctors, Admin, Pharmacy, and Lab modules. Manages dynamic registrations, approvals, and order dispatch.

`frontend/lib/pdfGenerator.js` — Official Clinical PDF Generation Engine

Client-side vector engine utilizing jsPDF to compile official hospital diagnostic summaries complete with hospital letterhead, demographics, ABCDE scores, return code, and digital sign-off.

`frontend/components/ClinicalDashboard.jsx` — Clinical Lesion Assessment Card Component

Renders the lesion dermoscopy view, ABCDE criteria breakdown, severity level, and consultation sign-off modal in a hospital EHR aesthetic.

`frontend/components/BodySelector.jsx` — Anatomical Body Selector Component

Component providing rapid anatomical region selection (Head, Torso, Arms, Legs) for patient triage intake.

Key Technical Defense Points for University / Dean Presentation

- Real-world EHR Integration: Connects Patients, Attending Doctors, Central Pharmacy, and Pathology Lab in real time.
- Administrative Governance: Enforces strict Admin approval on all first-time sign-ups before clinical access is granted.
- Explainable AI (XAI): Diagnostic inference adheres strictly to the clinical ABCDE dermatological guidelines.
- Production Reliability: Fully validated Next.js Turbopack build with 19 static & dynamic routes and zero runtime errors.